

WORKSHEET - NAPLAN Year 9

Number Rates & Ratios

Teacher: Niné Labuschagne

Exam Equivalent Time: 13.75 minutes (based on NAPLAN allocation of ~1.25 min/mark)



Worked Solutions

1. Number, NAP-61529

Ratio 3 : 2

Let x = Number of Carly's dolls

$$\frac{x}{12} = \frac{2}{3}$$

$$\begin{aligned} x &= \frac{2 \times 12}{3} \\ &= 8 \text{ dolls} \end{aligned}$$

2. Number, NAP-79138

$$\begin{aligned} \text{Ratio} &= 40:600 \\ &= 1:15 \end{aligned}$$

3. Number, NAP-96744

Let x = litres of red paint needed.

$$\frac{5}{8} = \frac{140}{x}$$

$$5x = 140 \times 8$$

$$x = \frac{140 \times 8}{5}$$

$$= 224 \text{ litres}$$

4. Number, NAP-02683

Brendan visits to Perth:

2020, 2017, 2014, 2011

Mining Expo held in years:

2020, 2018, 2016, 2014

$\therefore$ 2014 was the most recent year.

5. Number, NAP-02685

$$\begin{aligned}\text{Ratio} &= 3:4 \\ &= 1.5:2\end{aligned}$$

$\therefore$ 1.5 cups of sugar, 2 cups of sultanas is
in the correct ratio.

6. Geometry, NAP-14547

$$\begin{aligned}\text{Distance} &= 5 \times 1500 \\ &= 7500 \text{ m} \\ &= 7.5 \text{ km}\end{aligned}$$

$$\begin{aligned}\therefore \text{Walk Time} &= \frac{7.5}{6} \\ &= 1.25 \text{ hours} \\ &= 1 \text{ hour } 15 \text{ minutes}\end{aligned}$$

7. Number, NAP-79167

$$\begin{aligned}\text{Paddock length} \\ &= 56 \times 0.8 \\ &= 44.8 \text{ m}\end{aligned}$$

$$\begin{aligned}\therefore \text{Number of Caroline's steps} \\ &= \frac{44.8}{0.7} \\ &= 64\end{aligned}$$

8. Number, NAP-79196

$$\begin{aligned}\text{Total eggs laid in one day} \\ &= 30\,000 \times 60 \times 24 \\ &= 4.32 \times 10^7\end{aligned}$$

9. Number, NAP-79194

$$\begin{aligned}\text{Females : Males} &= 13:11 \\ \therefore \text{In the room, there are} \\ &26 \text{ females and } 22 \text{ males.}\end{aligned}$$

$\therefore$ 4 females need to leave for the
ratio to become 1 : 1

10. Number, NAP-08307

$$\begin{aligned}\text{Usage per year of one globe} \\ &= 30 \times 52 \\ &= 1560\end{aligned}$$

$$\begin{aligned}\therefore \text{Time until replacement of globe} \\ &= \frac{10\,000}{1560} \\ &= 6.41\dots \\ &= 6 \text{ years (nearest year)}\end{aligned}$$

11. Number, NAP-55673

$$\text{Total millilitres} = \frac{475}{1.5} = 316.67\dots$$

$$\begin{aligned}\text{Sand flow} &= \frac{316.67\dots}{15} \\ &= 21.1 \text{ ml per minute}\end{aligned}$$